

Human-like Bangla Text-to-Speech Synthesis using the Orpheus TTS Model Fine-tuning

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Abstract—Bangla text-to-speech (TTS) synthesis faces persistent challenges due to limited speech resources and complex linguistic features. This paper presents a fine-tuned Orpheus neural TTS model aimed at generating human-like Bangla speech with higher naturalness and clarity. A multilingual Orpheus model was fine-tuned using an open-source Bangla speech dataset. The fine-tuned Orpheus model achieves a substantial improvement in speech quality, especially natural voice tone. The Orpheus fine-tuned Bangla TTS model shows exceptional performance with speaker similarity of 99.96%, Character Error Rate (CER) of 5.8%, Word Error Rate (WER) of 11.1%, along with high similarity in Mel-Cepstral Spectrum. The results show that the fine-tuned Orpheus TTS excels compared to other TTS systems. The framework put forth outlines a productive and scalable process for developing high-quality Bangla TTS systems and extends the effectiveness of Orpheus to more low-resource languages.

Keywords—Bangla TTS, Orpheus TTS model, Speech Synthesis, Fine-Tuning, Bengali Text-to-Speech.

I. INTRODUCTION

Text-to-Speech (TTS) technology has gone through remarkable advances in the previous decade with the arrival of deep neural networks and abundant datasets. Models like Tacotron, WaveNet, and other end-to-end models have reached quality that is comparable to human speech in several high-resource languages. But for less speech-resourced languages such as, Bangla (Bengali) reaching similar natural speech is an open challenge with either limited availability of data, insufficient high-quality corpora, and limited modeling metrics and speaker variability.

There are encouraging new works in Bangla TTS recently. As just a few examples, Advancing Bangla text-to-speech synthesis using a VITS-based model with a custom dataset and comprehensive evaluation reports a VITS-based architecture that is trained with about 20 hours of Bangla speech with high intelligibility and naturalness, surpassing previous systems [1]. Further, BnTTS: Few-Shot Speaker Adaptation in Low-Resource Setting suggests speaker adaptation along multilingual TTS pipelines for Bangla with a big speech corpus and obtaining excellent subjective measures even with small amounts of new speaker data [2].

By contrast, transfer learning and fine-tuning methods more generally have been successful in cross-speaker or low-resource situations. Transfer Learning Framework for Low-Resource Text-to-Speech using a Large-Scale Unlabeled Speech Corpus demonstrate that we can get good naturalness and intelligibility [3]. Further, in Adapting TTS models for

New Speakers using Transfer Learning, the authors demonstrate that fine-tuning an existing single-speaker TTS model with around 30 minutes of data for a new speaker can yield performance comparable to a model trained from scratch on many hours of data [4].

Orpheus TTS is a recently introduced architecture that has human-like voice synthesis with fine-tuning, emotion markings, and expressive prosody i.e. with specific tokens for non-verbal expressions. We propose a fine-tuned Orpheus TTS model that is fine-tuned on a hand-picked open-source Bangla speech dataset in this work.

II. RELATED WORK

Neural text-to-speech (TTS) research has made spectacular advances from autoregressive waveform synthesizers to end-to-end, parallel, and conditional generative approaches. Initial innovations like WaveNet proved that unconditional neural autoregressive waveform models can generate highly natural speech and inspired subsequent end-to-end acoustic and neural vocoder models [5]. End-to-end sequence-to-sequence acoustic models such as, Tacotron simplified the TTS pipelines by directly learning short-cut mappings from characters/phonemes to spectrograms with the capability to attain high naturalness with fewer hand-crafted modules. Such advances opened the door to fully single-stage TTS systems and to ventures in robust alignment, prosody modelling and expressive synthesis [6].

Latest single-stage, parallel methods (most notably VITS) put variational inference, normalizing flows and adversarial training together to give us end-to-end TTS that's both high quality and fast to sample. VITS and variants all model duration and latent prosodic factors explicitly, which makes them good baselines for high-quality low-latency synthesis and also a commonly used comparison point for new TTS architectures and fine-tuning approaches [7].

Bangla (Bengali) TTS is one such active low-resource research area. Later Bangla research builds upon them by targeting large-scale pretraining, few-shot speaker adaptation, and architecture customization, with an emphasis on text normalization, phoneme mapping, and prosody modeling for the phonology and orthography of Bangla [8].

A research paper with step-by-step guidance on building a TTS model using an available open-source dataset for low-resource languages, including Bangla and several other low-resource languages, was demonstrated [9].

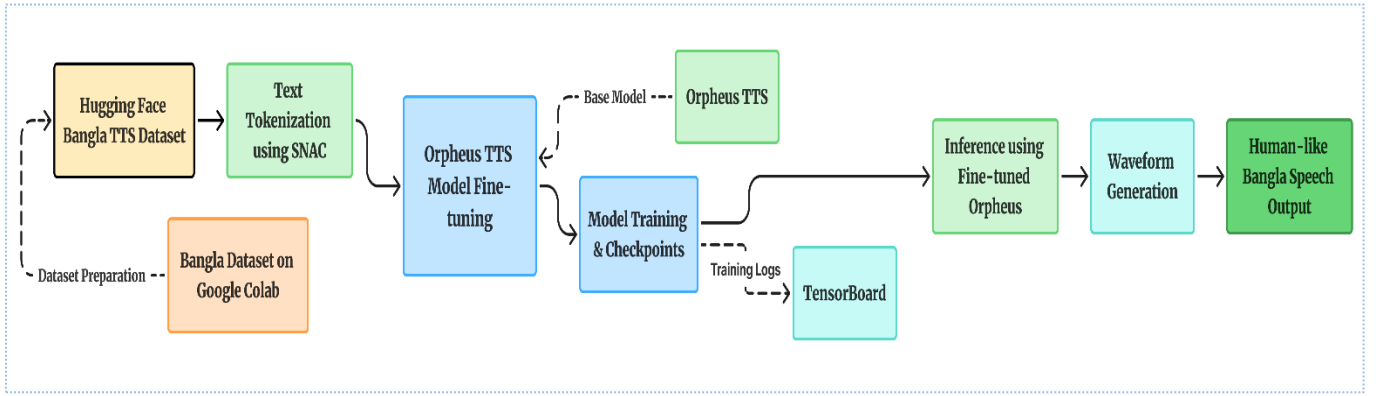


Fig. 1: Proposed System Overview

Orpheus TTS is a modern speech model displaying a strong level of naturalness, advanced prosody controllability, and zero-shot voice cloning with the aid of extensive pretrained LLM architectures and tuning with speech data [10].

This paper addresses the gap by (i) detailing language-specific dataset preprocessing and fine-tuning of the state-of-the-art human sounding TTS model Orpheus for Bangla language and, (ii) reporting evaluation metrics (Mel-Cepstral Spectrum analysis, Speaker similarity, CER, WER) along with fine-tuning performance analysis.

III. METHODOLOGY

The proposed system aims to develop a natural and human-like Bangla Text-to-Speech (TTS) model through the fine-tuning of the advanced Orpheus neural TTS architecture. The methodology follows a systematic and data-driven pipeline designed to ensure linguistic accuracy, acoustic naturalness, and efficient model convergence.

Orpheus is a TTS system that uses a neural network, combining transformer and diffusion modules. It provides high-quality prosody and expressive synthesis in multiple languages. For the Bangla fine-tuning task, the model trains on a processed open-source Bangla speech dataset with high-quality recordings matched to their transcriptions. The dataset was preprocessed using Google Colab for compatibility with the Orpheus model. The fine-tuning of the TTS model has been done several times with different GPU accelerations to get the best output. Different epochs were used until the desired performance was achieved. After every successful fine-tuning, the model was tested for natural-sounding voice generation capability again and again.

The fine-tuning process focuses on improving the Orpheus model using the Bangla dataset. This involves adjusting the parameters of both the encoder and decoder to capture the phonetic, prosodic, and tonal features of the Bangla language. Periodic checkpoints are saved to track training progress. We use evaluation metrics like Mel-Cepstral Spectrum Analysis, Character Error Rate (CER), Word Error Rate (WER), and training loss performance to measure how well the model is performing. After training is finished, the inference pipeline translates new Bangla text into natural-sounding audio clips. The final output is a high-

quality, human-like Bangla speech waveform that demonstrates correct pronunciation, natural prosody, and an expressive tone. All experimental stages from data preprocessing to fine-tuning and evaluation take place on Google Colab. This utilizes its strong GPU infrastructure and built-in monitoring tools like TensorBoard for training visualization.

The following are the study's main contributions:

1. A fine-tuning framework for the Orpheus model in Bangla,
2. Bangla open-source dataset processing for Orpheus TTS model fine-tuning.
3. Comprehensive testing with objective measures (e.g. Mel-Cepstral Spectrum Analysis, Character Error Rate, Word Error Rate) and fine-tuning performance analysis with training loss.
4. Relative evaluation indicating gains compared to baseline Bangla TTS systems in human-likeness, with relatively minimal data demands.

The overview of the proposed method of this study is depicted in Fig. 1, where the full system is illustrated, from the dataset processing to the human-like Bangla speech generations.

A. Data collection

For the present study, a publicly accessible open-source dataset for Bangla Text-to-Speech (TTS) was utilized [11]. This dataset underwent a process of refinement and was subsequently uploaded to the Hugging Face platform to facilitate fine-tuning [12]. The dataset is presented in Table I.

The corpus consists of nearly three hours (2 hours and 56 minutes) of acoustic data for the speech that has been gathered from six native speakers of Bangladesh. The complete dataset consists of 1,891 audio samples along with their respective Bangla transcription texts. Those audio files are all saved in the .wav format at a 48 kHz sampling rate, ensuring that the audio quality is top-notch and ready for the training of neural TTS models. The audio files were resampled to 24 kHz while fine-tuning. The dataset also includes a lexicon of 69,277 distinct Bangla words that offer rich phonetic and linguistic variety needed for the efficient model's generalization and prosody learning.

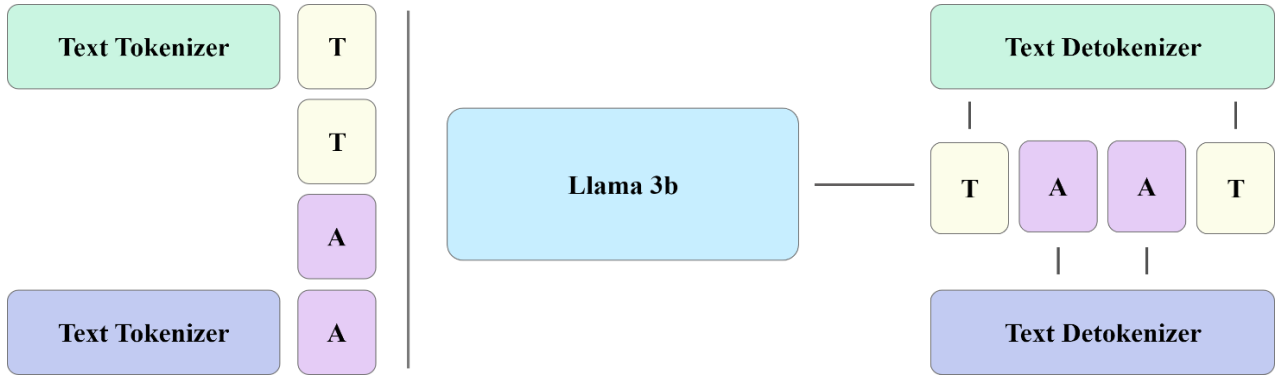


Fig. 2: Architecture of Orpheus TTS Model

TABLE I. Bangla TTS Dataset

Bangla Text-to-Speech Dataset	
Number of Speakers	6
Number of Audio Files	1,891
Audio Files Format	.wav
Audio Channels	1 (Mono)
Audio Sample Rate	48 kHz
Audio Sample Size	16 bit
Total Duration (Hours)	2:56:18
Speakers	Male
Transcriptions	Bangla (Bn)
Lexicon Entries	69,277

B. Orpheus TTS model architecture

The Orpheus TTS architecture [13] integrates text and speech tokenization into a unified multimodal framework built on top of a Llama 3b language model. Text input is first processed by the Text Tokenizer, while the corresponding speech data is encoded by the Speech Tokenizer. The token streams are then combined in the Llama 3b body for the acquisition of the detailed relationships between the acoustic and the linguistic representations. On the decoding end, the Speech Detokenizer and the Text Detokenizer are utilized for the conversion of the synthesized tokens back to understandable textual format and natural audio waveforms.

The Fig. 2 Shows the architecture of the Orpheus TTS model. This design enables bidirectional learning between text and speech, resulting in human-like voice generation. Unlike traditional TTS models, Orpheus aligns text and speech tokens together. Orpheus also supports low-resource adaptation.

C. Fine-tuning Setup

The Orpheus TTS model for Bangla speech synthesis was fine-tuned on Google Colab Notebook. For the GPU setup, we used both the NVIDIA T4 and the NVIDIA A100-SXM4-40GB GPU. We also used the Unsloth framework, which provided up to twice the speed of traditional methods.

IV. RESULTS AND DISCUSSION

The performance of the fine-tuned Orpheus Bangla TTS model was evaluated to assess its capability to generate natural and intelligible Bangla speech. The evaluation metrics considered include Training performance of fine-tuning process, Mel-Cepstral Spectrum analysis, Speaker Similarity, Character Error Rate (CER) and, Word Error Rate (WER). These metrics collectively measure spectral accuracy, linguistic clarity, and perceptual quality.

i. Fine-tuning performance

The fine-tuning performance of the Orpheus TTS model for Bangla is depicted in Fig. 3.

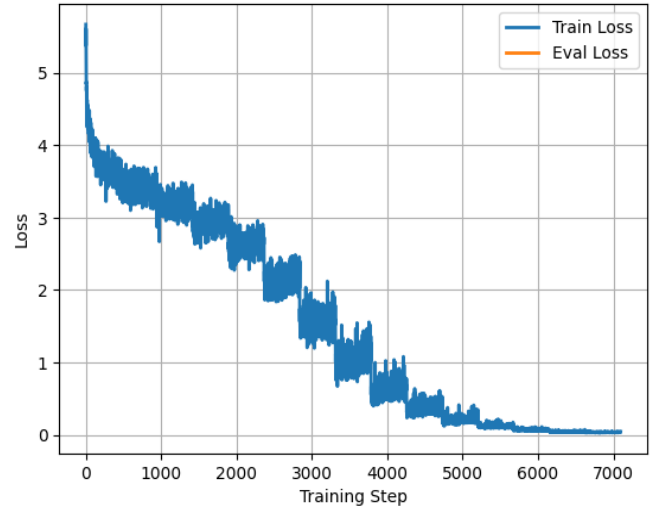


Fig. 3: Steps vs. Training loss of fine-tuned Orpheus TTS model

The extended fine-tuning of the Orpheus Bangla Text-to-Speech (TTS) model over 7,095 steps shows a strong performance while the training loss decreased from 5.667 to as little as 0.045. This iterative optimization is the proof for the model's improved performance. The training involved 15 epochs, completing a total of 7,095 optimization steps. To improve both efficiency and performance in learning, only 97.26 million out of 3.39 billion parameters, which is 2.86%, were set as trainable during fine-tuning. The final fine-tuning of 15 epochs took a total of 153.25 minutes on the NVIDIA A100 GPU on the Google Colab Notebook.

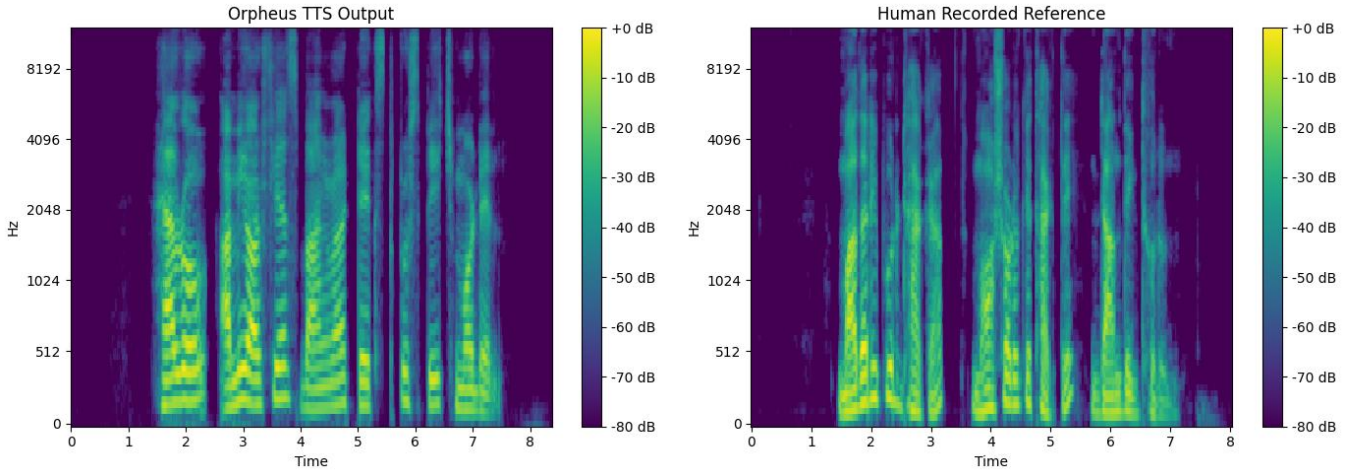


Fig. 4: Mel-Cepstral Spectrum Comparison Between Fine-Tuned Bangla Orpheus TTS Output and Human Speech

As a result, the steady drop in training loss is more than just a statistical gain, it leads to clear improvements in the synthesized speech, marked by greater naturalness, better clarity, and fewer noticeable artifacts.

ii. Mel-Cepstral Spectrum

Based on Fig. 4, which illustrates the Mel-Cepstral Spectrum analysis, the spectrum visuals are quite similar between the Orpheus TTS model-generated speech and the human recorder voice. This indicated that the generated output from the fine-tuned Orpheus Bangla TTS model was quite human-like.

iii. Speaker Similarity of fine-tuned Bangla TTS

The fine-tuned Orpheus Bangla TTS demonstrated very high speaker similarity with the original speaker of the recorded voice clip. This result is the evidence that Orpheus Bangla TTS is better in speech generation. The speaker similarity value is 99.96%. Fig. 5 shows the snippet of the experiment where an audio clip with a long Bangla sentence was chosen for the SNAC (Self-supervised Neural Audio Codec) based speaker similarity evaluation.

◆ SNAC Speaker Similarity Scores:
TTS_ban_00737_00530142458.wav: 0.9996

▶ 0:00 / 0:11 🔊 ⋮

✅ Speaker similarity evaluation complete!

Fig. 5: Speaker Similarity of Proposed TTS

iv. Character Error Rate (CER) of Orpheus TTS

To evaluate the fine-tuned Orpheus model, the Character Error Rate (CER) and Word Error Rate (WER) were calculated. For the first generated audio, the CER was as perfect as zero. For the second generated audio, the CER was 11.1%. So, the mean CER was 5.8% for two randomly generated speeches, which indicates higher performance. The lower CER indicates the fine-tuned model has a lower error rate per character of the speech generation. The open-source Bangla ASR model [14], fine-tuned on the Whisper

model, was utilized for both CER and WER evaluation. The mean CER evaluation is presented in Fig. 6.

◆ Audio 1: output_1_tts.wav

▶ 0:00 / 0:04 🔊 ⋮

Reference : গুগল বর্তমান বিশ্বের সবচেয়ে জনপ্রিয় সার্চ ইঞ্জিন।
Hypothesis: গুগল বর্তমান বিশ্বের সবচেয়ে জনপ্রিয় সার্চ ইঞ্জিন।
CER : 0.000

◆ Audio 2: output_2_tts.wav

▶ 0:00 / 0:04 🔊 ⋮

Reference : ডাচ বাংলা ব্যাংক সর্বাধিক এটিএম বুথ নেটওয়ার্ক পরিচালনা করে।
Hypothesis: ডাচ বাঙ্গ ব্যাংক সরলিক এটিএম বুথ নেটওয়ার্ক পরিচালনা করে।
CER : 0.117

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Mean CER over 2 audios: 0.058

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Fig. 6: CER of Proposed TTS

v. Word Error Rate (WER) of Orpheus TTS

The WER for a randomly generated speech was as less as 11.1%. Here, the highest WER was found 22.2% for the second sentence. The mean WER evaluation is demonstrated in the Fig. 7.

◆ Audio 1: output_1_tts.wav

▶ 0:00 / 0:04 🔊 ⋮

Reference Text: গুগল বর্তমান বিশ্বের সবচেয়ে জনপ্রিয় সার্চ ইঞ্জিন।
Hypothesis Text: গুগল বর্তমান বিশ্বের সবচেয়ে জনপ্রিয় সার্চ ইঞ্জিন।
WER: 0.000

◆ Audio 2: output_2_tts.wav

▶ 0:00 / 0:04 🔊 ⋮

Reference Text: ডাচ বাংলা ব্যাংক সর্বাধিক এটিএম বুথ নেটওয়ার্ক পরিচালনা করে।
Hypothesis Text: ডাচ বাঙ্গ ব্যাংক সরলিক এটিএম বুথ নেটওয়ার্ক পরিচালনা করে।
WER: 0.222

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Mean WER over 2 audios: 0.111

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Fig. 7: WER of Proposed TTS

vi. Comparisons with fine-tuned Orpheus TTS

The WER (Word Error Rate) comparison among Bangla TTS models with fine-tuned Orpheus Bangla TTS model is presented in Fig. 8.

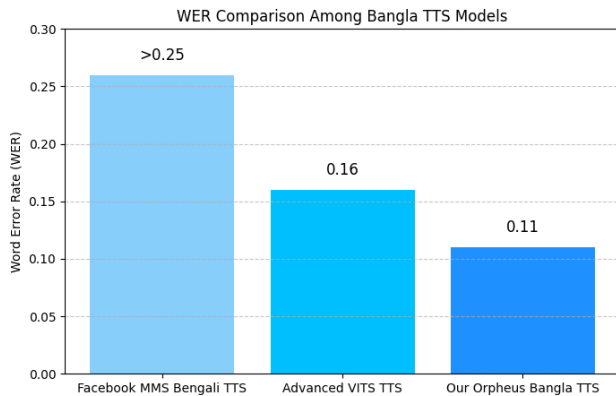


Fig. 8: WER Comparison with Proposed TTS model

Here, as presented in the graph, a lower value of WER indicated better speech synthesis capability. Our proposed fine-tuned Orpheus Bangla TTS demonstrates better performance than other VITS-based TTS models, such as Facebook's MMS Bengali TTS model [15] and the Advanced VITS TTS model [1].

V. CONCLUSION

In this research, we developed and refined a high-quality Bangla text-to-speech (TTS) model, Orpheus-3B-TTS-Bangla-V4, with speaker similarity as high as 99.96%, using transformer-based architectures and efficient training techniques. The system was trained with the Unsloth framework. This setup allowed for faster fine-tuning with minimal computational demands while only training 2.86% of the total parameters. The proposed model performed well in generating natural, clear, and engaging Bangla speech. Its design captured the phonetic and prosodic details of the Bangla language. It tackled challenges such as pronunciation accuracy and contextual fluency.

The results show that the lightweight fine-tuning approach works well. It made notable improvements while maintaining good computational efficiency. The use of optimized training parameters and balanced data also ensured high-quality audio synthesis for different text inputs. This research also contributed to the progress of the field of natural language processing research in the Bangla language. Future work will look at expanding the dataset to include emotional expressions and multi-speaker features to enhance naturalness and versatility. Overall, this study significantly contributes to Bangla speech technology by providing a scalable, high-quality, and resource-efficient solution for generating natural Bangla speech.

REFERENCES

[1] S. Kumar, S. Kumar, K. Sathe, and J. Pati, "Advancing Bangla text-to-speech synthesis using a VITS-based model with a custom dataset and comprehensive evaluation," *Discover Computing*, vol. 28, no. 1, pp. 1–23, Dec. 2025, doi: 10.1007/S10791-025-09701-3/FIGURES/8.

[2] M. J. I. Basher *et al.*, "BnTTS: Few-Shot Speaker Adaptation in Low-Resource Setting," Feb. 2025, Accessed: Oct. 12, 2025. [Online]. Available: <https://arxiv.org/pdf/2502.05729>

[3] M. Kim, M. Jeong, B. J. Choi, S. Ahn, J. Y. Lee, and N. S. Kim, "Transfer Learning Framework for Low-Resource Text-to-Speech using a Large-Scale Unlabeled Speech Corpus," *Proceedings of the Annual Conference of the International Speech Communication Association, INTERSPEECH*, vol. 2022-September, pp. 788–792, Oct. 2022, doi: 10.21437/Interspeech.2022-225.

[4] P. Neekhara, J. Li, B. Ginsburg, P. Neekhara, J. Li, and B. Ginsburg, "Adapting TTS models For New Speakers using Transfer Learning," *ArXiv*, p. arXiv:2110.05798, Apr. 2021, doi: 10.48550/ARXIV.2110.05798.

[5] A. van den Oord *et al.*, "WaveNet: A Generative Model for Raw Audio," Sep. 2016, Accessed: Oct. 14, 2025. [Online]. Available: <https://arxiv.org/pdf/1609.03499>

[6] Y. Wang *et al.*, "Tacotron: Towards End-to-End Speech Synthesis," *Proceedings of the Annual Conference of the International Speech Communication Association, INTERSPEECH*, vol. 2017-August, pp. 4006–4010, Mar. 2017, doi: 10.21437/Interspeech.2017-1452.

[7] J. Kim, J. Kong, and J. Son, "Conditional Variational Autoencoder with Adversarial Learning for End-to-End Text-to-Speech," 2021, Accessed: Oct. 14, 2025. [Online]. Available: <https://jaywalnut310.github.io/vits-demo/index.html>

[8] P. Bhattacharjee, R. S. Raju, A. Ahmad, and M. S. Rahman, "End to End Bangla Speech Synthesis," *2021 International Conference on Science and Contemporary Technologies, ICSCCT 2021*, Aug. 2021, doi: 10.1109/ICSCCT53883.2021.9642672.

[9] K. Sodimana *et al.*, "A Step-by-Step Process for Building TTS Voices Using Open Source Data and Frameworks for Bangla, Javanese, Khmer, Nepali, Sinhala, and Sundanese," in *6th Workshop on Spoken Language Technologies for Under-Resourced Languages, SLTU 2018*, The International Society for Computers and Their Applications (ISCA), 2018, pp. 66–70, doi: 10.21437/SLTU.2018-14.

[10] "canopyai/Orpheus-TTS: Towards Human-Sounding Speech." Accessed: Oct. 14, 2025. [Online]. Available: <https://github.com/canopyai/Orpheus-TTS>

[11] "openslr.org." Accessed: Nov. 08, 2025. [Online]. Available: <https://www.openslr.org/37/>

[12] "utshobs/bangla-tts · Datasets at Hugging Face." Accessed: Nov. 08, 2025. [Online]. Available: <https://huggingface.co/datasets/utshobs/bangla-tts>

[13] "Canopy Labs." Accessed: Nov. 08, 2025. [Online]. Available: https://canopylabs.ai/releases/towards_human_sounding_tts

[14] "bangla-speech-processing/BanglaASR · Hugging Face." Accessed: Nov. 08, 2025. [Online]. Available: <https://huggingface.co/bangla-speech-processing/BanglaASR>

[15] "facebook/mms-tts-ben · Hugging Face." Accessed: Nov. 08, 2025. [Online]. Available: <https://huggingface.co/facebook/mms-tts-ben>